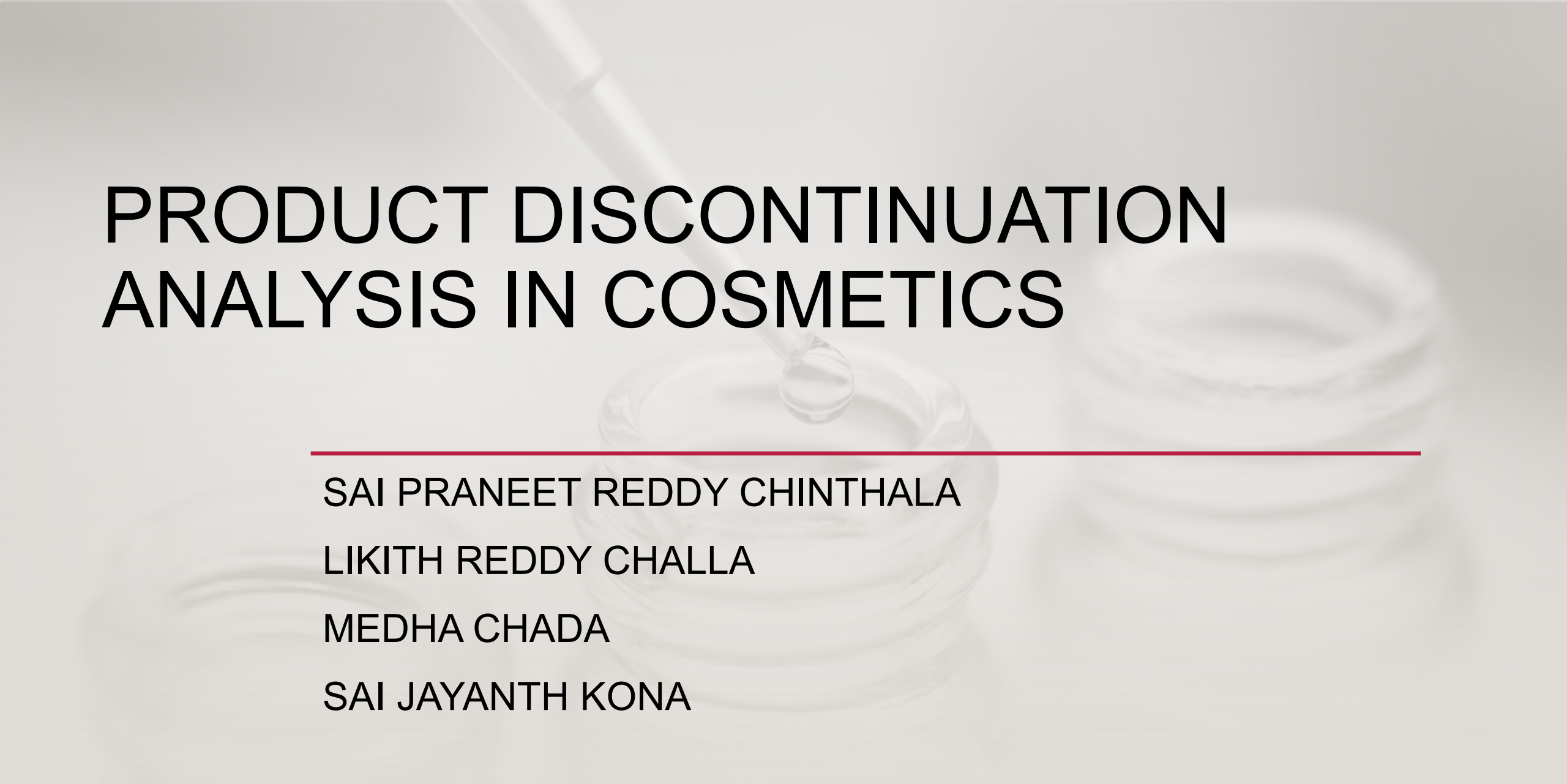


PRODUCT DISCONTINUATION ANALYSIS IN COSMETICS

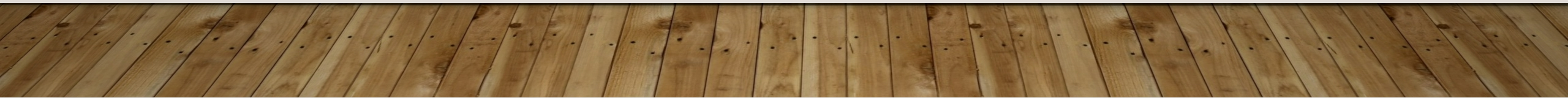


SAI PRANEET REDDY CHINTHALA

LIKITH REDDY CHALLA

MEDHA CHADA

SAI JAYANTH KONA



PROBLEM DESCRIPTION

- The Cosmetic industry
- The Johnson & Johnson Case
- Research questions
 - Can we classify products based on their likelihood of discontinuation?
 - Detecting trends found in cosmetic formulations among different brands

DATA DESCRIPTION

- The chosen Dataset spans over 114,635 rows containing details of manufacturers and chemical compositions.
- The key variables are Chemical count, Products per brand and chemicals per category.

MODELLING APPROACH

- Logistic Regression and Random Forest were used to classify products based on their likelihood of discontinuation and predict if a product will be discontinued.
- K-means Clustering was used to identify trends in brand products and chemical count.

RESULTS

- Top 3 Categories present in Primary category that affect Discontinuation are Personal Care Products, Baby Products, Non-Coloring Hair Care Products.
- Higher chemical count relates to a higher discontinuation rate.
- Both Logistic Regression and RandomForest have an accuracy of 87%
- 2 different clusters were formed where cluster 1 refers to products with simple formulations and cluster 2 refers to products with complex formulations.

MANAGERIAL IMPLICATIONS

- Simplify ingredients in high-risk products to reduce discontinuation probability.
- Identify alternative ingredients for products with discontinuation risk.
- This gives high ingredient brands to promote their products based into luxury while the low ingredient ones can market theirs around simple beauty.

LIMITATIONS

- The model performs well in predicting if a product will not be discontinued but faces issues vice-versa.
- Absence of Market Trends-Data on sales, pricing, or consumer demand, which affect discontinuation, are not available.